HYDROGEN TRACKER

A centralized view of hydrogen operations

- Production • Storage • Shipments • Deliveries • Analytics

The Problem

Hydrogen workflows can become fragmented across teams and tools.

- Production, tank inventory, shipments, and customer deliveries are connected but often tracked separately.
- Manual coordination can make it difficult to maintain a consistent operational picture.
- Teams need clearer visibility into records, movement status, and exceptions.

Who Is Affected?

A shared operational view for the people coordinating hydrogen supply.

- Plant and production operators maintaining production records.
- Storage and logistics teams monitoring tank and shipment information.
- Dispatch coordinators and customer-facing teams following delivery status.

Our Solution

Hydrogen Tracker brings key workflows into one web application.

- Role-oriented dashboards and authenticated access.
- Production and storage records, including an Add Tank workflow.
- Shipment, delivery, and customer management.
- Map visualization, charts, and notifications where configured.

Core User Journey

From production records to delivery visibility.

- Sign in and review the dashboard.
- Create or review production and tank records.
- Follow shipments and deliveries; maintain customer information.
- Use map and analytics views to understand the available operational data.

Technology Overview

MERN-style full-stack architecture (confirm exact packages in the repository).

- React + Vite: user interface and page navigation.
- Node.js + Express: API routes and application logic.
- MongoDB: persistence for application records.
- Socket.IO / chart / map components: use where implemented and configured.

Value & Scope

Designed for operational visibility—not industrial process control.

- Centralizes related records in a common interface.
- Helps teams follow tank, shipment, and delivery information.
- Provides visual context through maps, charts, and notifications.
- Prototype data and workflows must be validated before real operational use.

Demo & Next Steps

Demonstrate the current build with sample data.

- Show dashboard → Add Tank → shipments/deliveries → map → analytics.
- Record a walkthrough of no more than three minutes.
- Add the public demo link to README.md after uploading the recording.
- Repository: github.com/shanmukhapriyab/hydrogentrackernew